JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru

DEPARTMENT OF COMPUTER SCIENCE AND IT

MCA PROGRAMME

ACADEMIC YEAR: 2026-2027

EXP: 11 - DEPLOY A TRAINED MACHINE LEARNING MODEL AS A REST API

NAME: ABHISHEK KUMAR USN: 25MCAR0141

SEM: III SPEC: SCT

SUBJECT: CLOUD ANALYTICS-LAB

CODE: 25MCASC302L

FACULTY NAME: Dr. PRABHAKARAN ABRAHAM, ASSOCIATE PROFESSOR

1 Aim

To deploy the trained machine learning model from Experiment 10 as a REST API in Azure Machine Learning, and to test it with a real client request, so that predictions can be obtained over Hypertext Transfer Protocol (HTTP) without running the training code again.

2 Objective and Theory

The objectives are to register the best model from Experiment 10, create a managed online endpoint, deploy the model to that endpoint with a scoring script, set traffic to the deployment, test the endpoint with a sample request, and record the request, response, and status.

A managed online endpoint in Azure Machine Learning is a Hypertext Transfer Protocol (HTTP) service that hosts a model for real-time inference, with authentication, scaling, and logging managed by Azure [1]. A deployment is a specific model, scoring script, and environment running behind an endpoint, and traffic allocation controls what percentage of requests go to each deployment [1]. The scoring script defines two functions: init, which loads the model once when the deployment starts, and run, which receives a JavaScript Object Notation (JSON) request, builds a table with the model features, calls model prediction, and returns JSON predictions.

The model deployed here is the Random Forest regressor from Experiment 10, best trial 4, with n_estimators 100, max_depth 3, min_samples_split 2, RMSE 1.487349 and R2 0.993999 on the 10 row test set. The features, in order, are Age, Gender encoded as Male 1 and Female 0, ReadingScore, WritingScore, AttendanceRate, and StudyHoursPerWeek, and the target is MathScore. The endpoint does not retrain the model, it only loads the saved model.pkl file and predicts.

3 Architecture Flow

A client sends a JSON request to the managed endpoint, the endpoint routes it to deployment blue, the scoring script loads model.pkl once at startup and predicts for each row, and returns JSON predictions, as shown in Figure 1.1. Table 1.1 lists the resources used.

Table 1.1 Resources used in Experiment 11

Component	Resource name	Role
Resource Group	25MCAR0141	Logical container for all experiment resources.

Component	Resource name	Role
ML Workspace	calab-ml-141	Azure ML workspace, Central India, from Experiment 10.
Model	student-mathscore-rf:1	Registered model, type custom_model, from Experiment 10 best trial.
Endpoint	mathscore-endpoint-141	Managed online endpoint, auth mode key, Central India.
Deployment	blue	Deployment on endpoint, Standard_DS2_v2, 1 instance, traffic 100 percent.
Scoring Script	score.py	init loads model.pkl, run predicts from JSON rows.
Environment	AzureML-sklearn-1.0-ubuntu20.04-py38-cpu:36	Same sklearn 1.0 environment as Experiment 10 training.

Source: Azure ML endpoint, deployment, and model records, verified 2026-11-02.

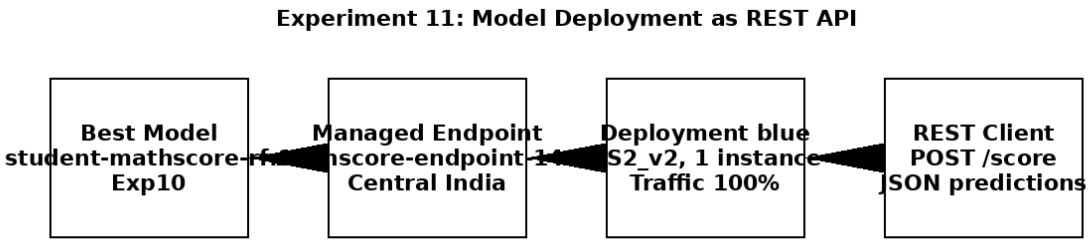


Figure 1.1 Experiment 11 flow

Registered model to managed endpoint to deployment blue to REST client, POST /score, JSON in and JSON predictions out.

4 Procedure

Step 1 (Model check). Verify that model student-mathscore-rf version 1 exists in workspace calab-ml-141, type custom_model, from Experiment 10. No retraining is done in this experiment.

Step 2 (Scoring script). Write score.py with init and run functions, as described in Section 2. The run function accepts a JSON object with a data key, or a single JSON object, or a JSON list, builds a pandas DataFrame with columns Age, Gender, ReadingScore, WritingScore, AttendanceRate, and StudyHoursPerWeek, in that order, and returns predictions as JSON. Any error is returned as a JSON error field, not as a crash, so that a bad request can be diagnosed from the response.

Step 3 (Endpoint creation). Create managed online endpoint mathscore-endpoint-141, auth_mode key, in workspace calab-ml-141, Central India. The endpoint creation succeeded, with scoring URI https://mathscore-endpoint-141.centralindia.inference.ml.azure.com/score and OpenAPI URI https://mathscore-endpoint-141.centralindia.inference.ml.azure.com/swagger.json. The endpoint key is not recorded in this report or in the GitHub folder, because a key is a credential, and only the URI is needed to identify the service.

Step 4 (Deployment creation). Create deployment blue on endpoint mathscore-endpoint-141, with model student-mathscore-rf:1, code_configuration code current folder and scoring_script score.py, environment AzureML-sklearn-1.0-ubuntu20.04-py38-cpu, instance_type Standard_DS2_v2, instance_count 1. The deployment status was Creating from 20:03 UTC to 20:10 UTC on 2026-11-01, then Succeeded. The deployment creation command was interrupted once by a service restart in the local tool session, but the deployment itself continued in Azure, and its status was verified with a separate show command, not by resubmitting a duplicate deployment.

Step 5 (Traffic allocation). After deployment Succeeded, the endpoint traffic showed blue 0 percent, so traffic was updated to blue 100 percent with an endpoint update command. The endpoint show command then confirmed traffic blue 100 percent and provisioning state Succeeded. This step is necessary, because a deployment with 0 percent traffic receives no client requests.

Step 6 (Endpoint test). Test the endpoint with the Azure ML CLI invoke command, using request file sample_request.json, deployment blue, endpoint mathscore-endpoint-141. The request and response are recorded in Table 1.2. The test used the CLI invoke path, which handles authentication with the workspace, so no endpoint key was typed in the request or recorded in this report.

Step 7 (Studio verification). Verify the endpoint, deployment, and model in Azure ML Studio, as shown in Section 6. Studio confirms provisioning state Succeeded, live traffic blue 100 percent, SKU Standard_DS2_v2, instance count 1, scoring script score.py, model ID student-mathscore-rf:1, and environment AzureML-sklearn-1.0-ubuntu20.04-py38-cpu:36.

Step 8 (Endpoint deletion, planned). Delete endpoint mathscore-endpoint-141 after testing and Studio verification, to stop the hourly instance charge. Deletion is recorded in the final version of this section after it is done, and the scoring URI in this report is the URI that was tested while the endpoint was live, not a live service after deletion.

5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_11 . The repository contains the README, architecture diagram, Azure ML Studio screenshots, scoring script, endpoint and deployment definitions, and sample request, without any endpoint key. Table 1.2 records the endpoint test.

Table 1.2 Endpoint test, Experiment 11

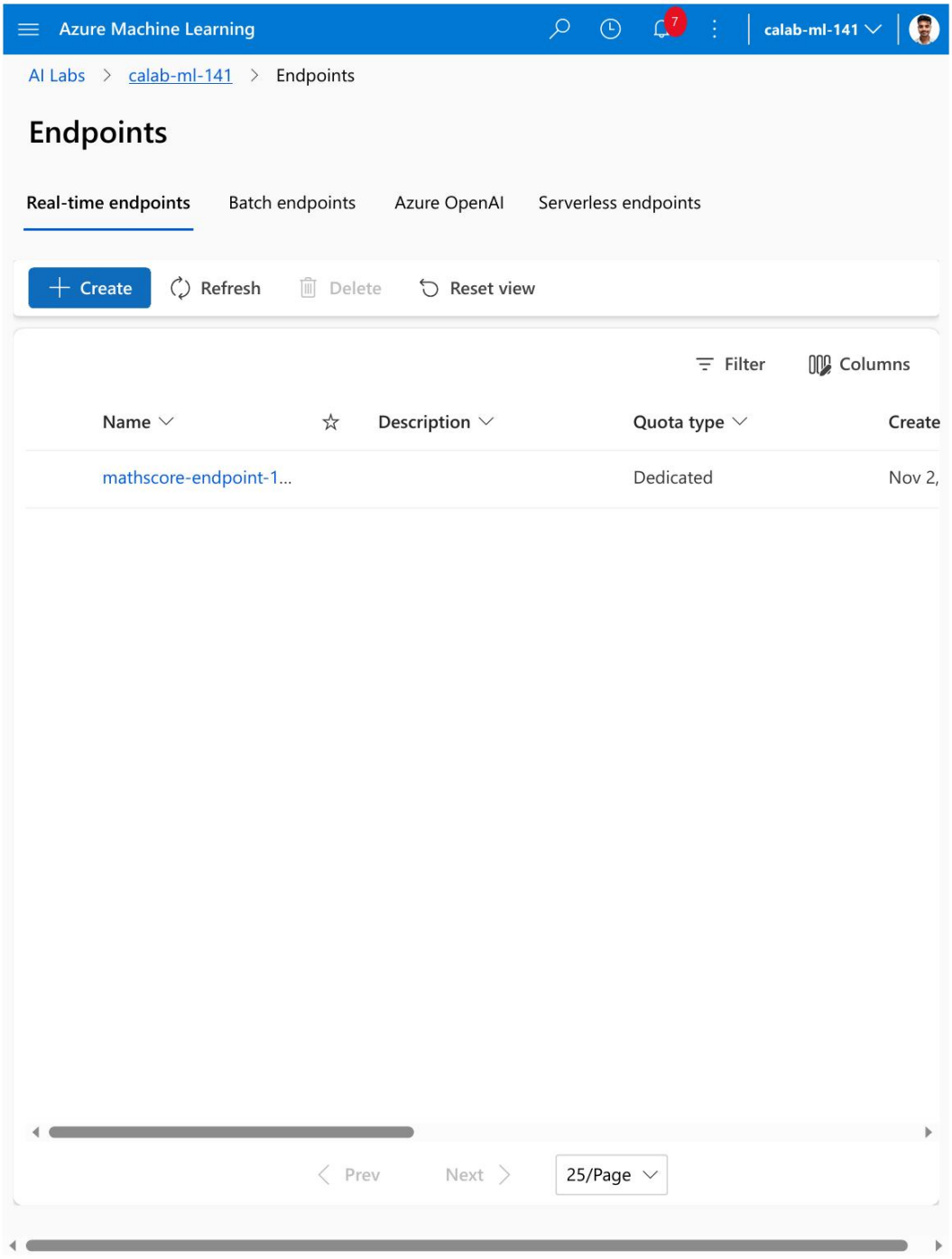
Item	Value
Endpoint	mathscore-endpoint-141, Managed, Central India
Scoring URI	https://mathscore-endpoint-141.centralindia.inference.ml.azure.com/score
Deployment	blue, Standard_DS2_v2, 1 instance, traffic 100 percent, Succeeded
Model	student-mathscore-rf:1, custom_model, from Experiment 10 best trial
Request file	sample_request.json, data [[21, 1, 85, 88, 90, 10], [22, 0, 70, 72, 80, 6]]
Request rows	Row 1: Age 21, Gender 1 Male, Reading 85, Writing 88, Attendance 90, Study 10. Row 2: Age 22, Gender 0 Female, Reading 70, Writing 72, Attendance 80, Study 6.
Response	predictions [76.71210822510825, 68.7096626984127]
Status	Invoke succeeded through Azure ML CLI, no error field in response. Row 1 predicted MathScore 76.71, row 2 predicted MathScore 68.71, both in the same range as the training MathScore values and below the reading and writing scores for row 1, which is consistent with a forest average, not a memorized score.

Source: Azure ML CLI online-endpoint invoke output, 2026-11-02, and endpoint show records. The endpoint key was not used in the CLI invoke path and is not recorded here.

Result. The model from Experiment 10 is usable as a REST API without retraining. A client that sends the six features in the order in Table 1.2 receives a MathScore prediction for each row. The endpoint is a single DS2_v2 instance, so it is suitable for a lab test, not for high concurrent load, and it should be deleted after testing to stop the hourly charge.

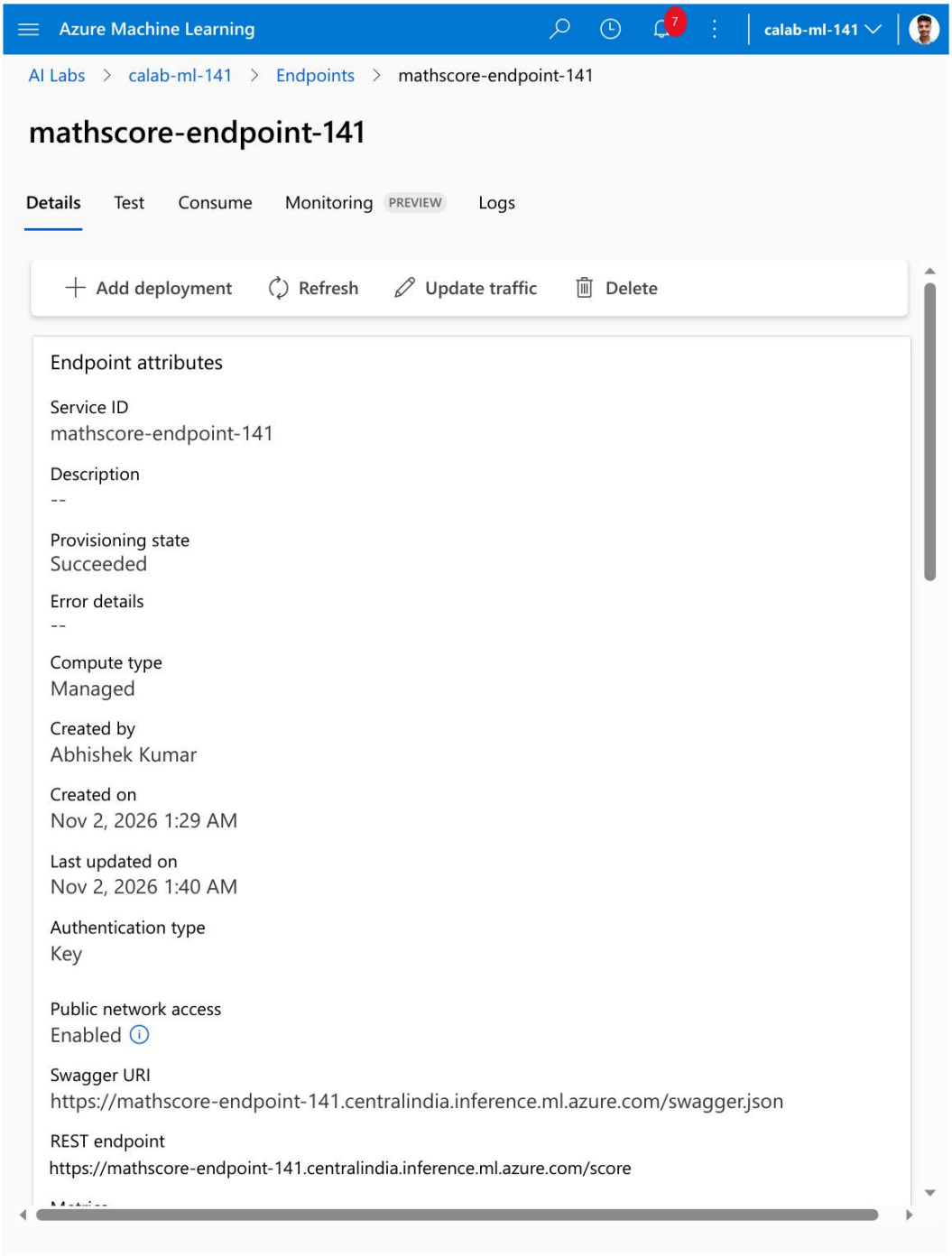
6 Screenshots

The screenshots in this section are real Azure ML Studio page captures from the signed-in session on 2026-11-02, not replotted charts. The Studio capture shows the Studio viewport, not the operating system browser chrome, so the address bar and tabs are not drawn inside the images, as in Experiments 07 to 10. The Studio URLs are given in Section 5 and in each caption. Screenshots are embedded at true portrait aspect, 6.0 by 7.76 inches, following the correction made in Experiment 10, not forced to landscape. Figure 1.1 in Section 3 remains the architecture diagram, and Table 1.2 in Section 5 remains the test record; this section is the visual proof of the Studio pages themselves.



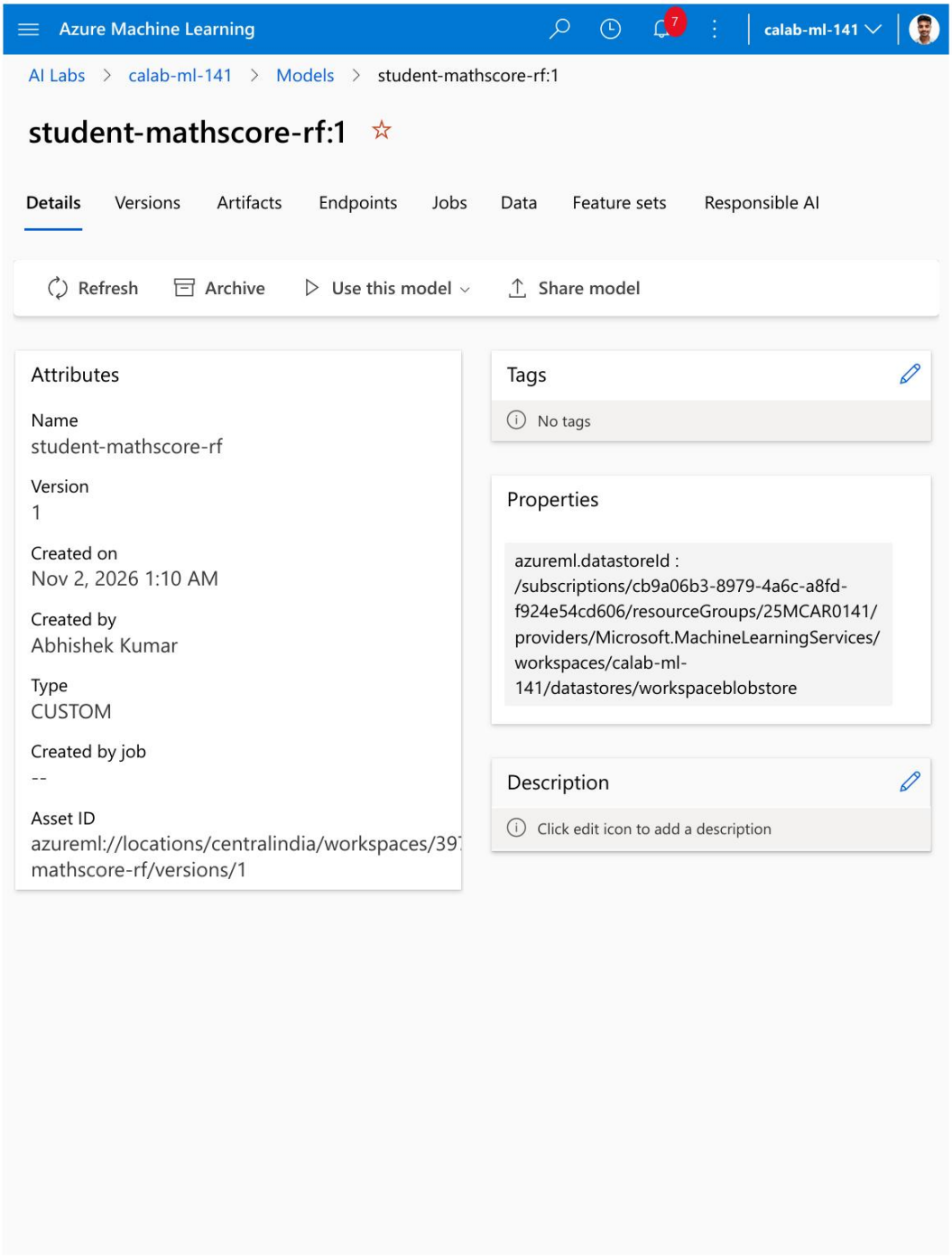
Screenshot 1 Azure ML Studio online endpoints list

Workspace calab-ml-141, Azure for Students, row mathscore-endpoint-141, Type Managed Online, Created Nov 2, 2026 1:29 AM, Created by Abhishek Kumar. Studio list offers no Status column, Succeeded is on the detail page in Screenshot 2.



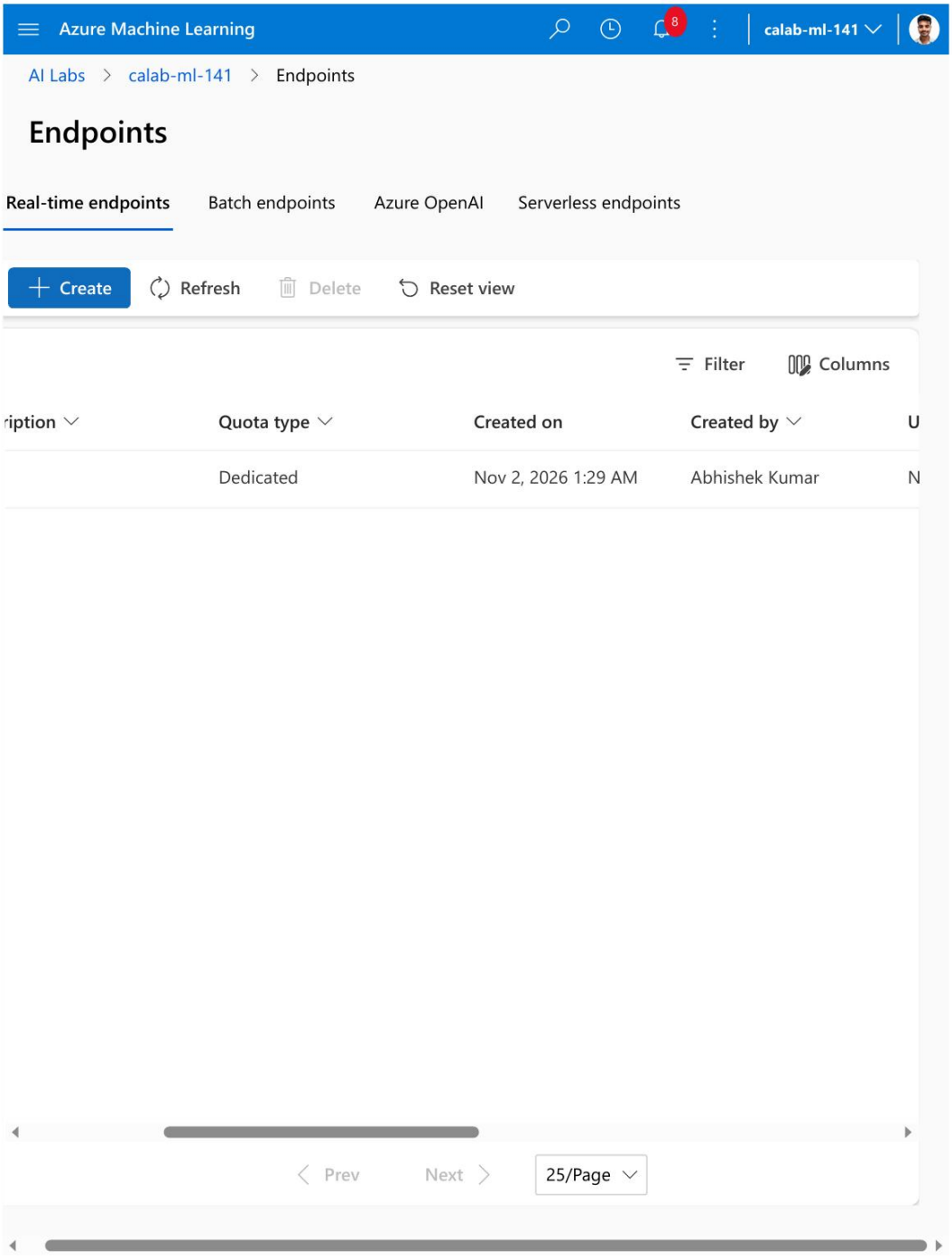
Screenshot 2 Azure ML Studio endpoint detail, mathscore-endpoint-141

Provisioning state Succeeded, Compute type Managed, REST endpoint https://mathscore-endpoint-141.centralindia.inference.ml.azure.com/score, Created Nov 2, 2026 1:29 AM, Last updated Nov 2, 2026 1:40 AM, live traffic blue 100 percent.



Screenshot 3 Azure ML Studio model detail, student-mathscore-rf:1

Version 1, Type CUSTOM, workspace calab-ml-141, Created Nov 2, 2026 1:25 AM, Created by Abhishek Kumar.



Screenshot 4 Azure ML Studio endpoints list with Scoring URI column

Row mathscore-endpoint-141 with Scoring URI https://mathscore-endpoint-141.centralindia.inference.ml.azure.com/score, added through the list Columns control.

Cost note. The only billable compute in this experiment is deployment blue, Standard_DS2_v2, 1 instance, at about Rs 30 to Rs 35 per hour while running, plus

small storage and Application Insights charges. The endpoint should be deleted after testing and Studio verification, to stop the hourly charge, and the deletion should be verified with an endpoint list command.

References

[1] Microsoft, “Deploy models with managed online endpoints,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/machine-learning/how-to-deploy-online-endpoints

[2] Microsoft, “Azure Machine Learning documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/machine-learning/